IAEA Coordinated Research Project on Development of Harmonized QA/QC Procedures for Maintenance and Repair of Nuclear Instruments

Introduction to the Test Procedures for Radiation Detectors and Associated Nuclear Modules Employed in Classical Detection Chains

PROCEDURE Nº MRNI-500
REV. D0

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1.- OBJECTIVE AND SCOPE

1.1.- Objective

The objective of this procedure is to describe the overall goals an strategies employed in the elaboration of the test procedures for radiation detectors and associated nuclear modules employed in classical detection chains to be used in Nuclear Instrumentation National Laboratories of Member States. These procedures were elaborated under the Coordinated Research Project "Development of Harmonized QA/QC Procedures for Maintenance and Repair of Nuclear Instruments" of the IAEA.

1.2.- Scope

This description is applicable to all the procedures included under the MRNI-5XX label number.

2.- NOTATION AND DEFINITIONS

2.1.- Notation

NIM	Nuclear Instrumentation Modules.
DMM	Digital Multimeter
MS	Member States
QA	Quality Assurance
QC	Quality Control
IAEA	International Atomic Energy Agency

2.2.- Definitions

2.2.1. Detector
A device that converts the energy of a photon or incident particle in an electric pulse.

3.- INTRODUCTION

The QA/QC test procedures for radiation detectors and associated nuclear modules were elaborated considering the use of common test tools such as:

- digital multimeters (DMM)
- NIM counter/timer
- frequency meter
- pulse generator
- power supplies (laboratory power supplies and high voltage power supplies as used to bias detectors)
- oscilloscope
- NIM crate with power supply

The procedures consider the opportunities and limitations encountered in MS laboratories in their regions. Therefore, this test procedures are focused on using basic and/or low cost test instruments which ensure that MS laboratories can follow all the provided advice on QA/QC procedures and/or test instructions.

3.1- Generalities

The procedures are numbered as follows:

-Procedure N° MRNI- 500 "Introduction to the Test Procedures for radiation detectors and associated nuclear modules employed in classical detection chains"
-Procedure N° MRNI-501 "Test procedure for Geiger-Mueller radiation detectors"
-Procedure N° MRNI-502 "Test procedure for Proportional Radiation Detectors"
-Procedure N° MRNI-503 "Test procedure for Ionization Chamber Radiation Detectors"
-Procedure N° MRNI-504 "Test procedure for Scintillation Radiation Detectors"
-Procedure N° MRNI-505 "Test procedure for Semiconductor Charged Particle Radiation Detectors"
-Procedure N° MRNI-506 "Test procedure for Silicon Lithium Drifted Radiation Detectors and Associated Electronics "
-Procedure N° MRNI-507 "Test procedure for High Purity Germanium Radiation Detectors and Associated Electronics"
-Procedure N° MRNI-508 "Test procedure for PIN diode radiation detectors"
-Procedure N° MRNI-509 "Test procedure for charge sensitive preamplifiers with feedback resistor"
-Procedure N° MRNI-510 "Test procedure for charge sensitive preamplifiers with optical feedback"
-Procedure N° MRNI-511 "Test procedure for spectroscopy amplifiers"
-Procedure N° MRNI-512 "Test procedure for single channel analyzers"
-Procedure N° MRNI-513 "Test procedure for counter systems and timers for nuclear measurements"
-Procedure N° MRNI-514 "Test procedure for Multichannel Analyzers Systems"

All the procedures have several common points in the Development section, as follows:

3.2.- Test Instruments
3.3- Test Conditions
3.3.1.- Background Radiation
3.3.2.- Temperature
3.3.3.- Radioactive source employed
3.4.- Test Circuits
3.5.- Measurements

Whenever possible, graphs and values, obtained experimentally with real equipment and detectors, are shown, trying to show results similar to the ones that are going to be obtained in the laboratory by applying the procedures.

There are several documents and didactic material elaborated by the IAEA, that can be used as support or complement about the theory of nuclear electronics, see the references: [5], [6], [7] and [8].

4.- ADMINISTRATION OF THE TEST REPORTS

4.1 Numbering of the reports.
All the generated technical test reports must have a unique and consecutive number.
4.2.- Personnel.
The test of detectors and electronic modules must be done by trained personnel.
4.3.- Technical Report of results.

4.3.1 The results of the test of detectors and electronic modules must be registered in a unique test report, stating the description of the detectors or/and electronic modules, mark, model, serial number, and all the test conditions, including the name of the person who made the tests.

4.3.2 All the test reports must be classified and keep in a folder for future consult.

5.- ACTION IN CASE OF NON CONFORMITIES

In case of nonconformities, an action has to be done and established in this paragraph.

6.- RESPONSIBILITIES

6.1.- Head of the Department.
Supervise that all the activities for testing of detectors and electronic modules follow the established procedure.

6.2.- Area Responsible.

6.2.1 Assure that all the electronic test equipment be in good operational conditions and calibrated.

6.3.2 Verify that all the activities for testing of detectors and electronic modules follow the established procedure

6.2.3 Verify that the test reports contain all the details of the testing of detectors and electronic modules.

6.2.4 Maintain a register and control of the test reports for all the detectors and electronic modules tested in the laboratory.

6.3 Operative Personnel.

6.3.1 Verify that all the electronic test equipment be in good operational conditions and calibrated.

6.3.2 Follow the steps established in this procedure for the testing of detectors and electronic modules.

6.3.3 Elaborate the test report of all the tests of the detectors and electronic modules.

6.3.4 Inform to the Area Responsible of any anomalous condition encountered during the test procedure.

7.- BIBLIOGRAPHY

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2.- ININ, "Elaboration of procedures and instructions" Procedure P.GC-12, Rev. January 13, 2004, ININ, MEXICO.

3.- IAEA "Quality Assurance for Safety in Nuclear Power Plants and other Nuclear Installations", Q1 Safety Guide, Safety Series N° 50-C/SG-Q, IAEA HQ, Vienna, Austria, 2001.

4.- EN ISO/IEC 17025:2000 IDT; ISO/IEC 17025:1999 IDT, General Requirements for the Competence of Calibration and Testing Laboratories, 1999.

5.- IAEA "Selected Topics on Nuclear Electronics" IAEA TECDOC 363, Vienna, 1987.

6.- IAEA "Troubleshooting in Nuclear Electronics" IAEA TECDOC 426, Vienna, 1987.

7.- IAEA "Nuclear Electronics Laboratory Manual" IAEA TECDOC 530, Vienna, 1987.

8.- IAEA "Nuclear electronics" didactic material developed under the IAEA project:"ICT-assisted training/learning materials in nuclear instrumentation" Vienna, 2004.

8.- ANNEXES

At least two annexes are included in the procedures:

Annex I . Flow Chart
Annex II. Test Report